

Question

In the following two variants of cubic ZnS , the metal atoms form alternating layered structures, with only one type of metal atom within the same layer, and they are all on the same plane. **A** – **F** represent six different elements.

In ABC_2 , **A** and **B** together occupy the Zn position in cubic ZnS , and there is no three-fold rotation axis in this crystal.

In DEF_2 , **D** and **E** together occupy the Zn position in cubic ZnS , and there is a three-fold rotation axis in this crystal.

How many atoms are there in a primitive unit cell of ABC_2 ? How many atoms are there in a structural motif?

How many atoms are there in a primitive unit cell of DEF_2 ? How many atoms are there in a structural motif?

A. 4; 4; 12; 12

B. 8; 4; 12; 12

C. 4; 4; 12; 4

D. 8; 4; 12; 4

E. 12; 12; 4; 4

F. 12; 12; 8; 4

G. 12; 4; 4; 4

H. 12; 4; 8; 4

I. All of the above options are incorrect.

Answer

Correct Answer: **C**

Detailed Explanation

In ABC_2 , **A** and **B** jointly occupy the position of **Zn** in the cubic ZnS . There is no threefold rotation axis in this crystal, so in ABC_2 it should be that **A** and **B** replace the two layers of **Zn** at $z = 0$ and $z = 0.5$ in the cubic ZnS unit cell with **Zn** as the vertex, instead of replacing the close-packed layers (which would lead to a threefold rotation axis).

CHECKPOINT

1 PTS

The substitution in ABC_2 is not the substitution of close-packed layers

The resulting unit cell is not a proper unit cell. Connecting the vertices and the face centers yields a proper unit cell, which should be a simple tetragonal lattice, containing 4 atoms, and the structural motif also contains 4 atoms.

CHECKPOINT

1 PTS

One proper unit cell of ABC_2 contains 4 atoms

CHECKPOINT

1 PTS

One structural motif of ABC_2 contains 4 atoms

In DEF_2 , **D** and **E** jointly occupy the position of Zn in the cubic ZnS. There is a threefold rotation axis in this crystal, so it is a substitution of close-packed layers.

CHECKPOINT

1 PTS

The substitution in DEF_2 is the substitution of close-packed layers

Using uppercase letters to represent the close-packed layers of Zn and lowercase letters to represent the close-packed layers of S, the stacking structure of cubic ZnS can be expressed as |AaBbCc|. After **D** and **E** alternately replace Zn in layers, it will become **Double subscripts: use braces to clarify**, which is a typical R-centered hexagonal lattice. A proper unit cell contains 12 atoms, which is 3 structural motifs, i.e., one structural motif contains 4 atoms.

CHECKPOINT

1 PTS

One proper unit cell of DEF_2 contains 12 atoms

CHECKPOINT

1 PTS

One structural motif of DEF_2 contains 4 atoms